Inexhaustible Energy Sources and Material Resources: Ontological Status as of July 2026

1. Ontological Framework of Planetary Resource Mechanics

The physical reality of the Earth is defined by its precise thermodynamic and spatial orientation within the solar system. Evaluated through the strict parameters of astrophysics, geophysics, and thermodynamics, the Earth functions as an open thermodynamic system with respect to energy and a closed thermodynamic system with respect to baryonic matter[1]. This ontological reality establishes the absolute baseline for planetary resources: energy continuously flows into the system from extraterrestrial sources (solar radiation) and from endogenous geological processes (radiogenic decay and primordial heat), while material atoms remain permanently bound within the planet's gravitational well, undergoing state changes rather than physical annihilation[1].

As of July 2026, empirical measurements, satellite telemetry, and applied physical models provide an exhaustive factual basis confirming the sheer inexhaustibility of the planet's primary energy and material resources relative to any existing or projected anthropogenic demand. This document presents a strict, non-evaluative constellation of facts regarding the presence, immutability, and quantification of these physical phenomena, alongside an analysis of the socio-economic mechanisms that construct artificial scarcity in an environment defined by physical abundance.

2. Solar Radiation: The Primary Extraterrestrial Energy Flux

2.1. Physical Constants and the Stefan-Boltzmann Application

Solar energy represents the primary exogenous energy input to the Earth system. The intensity of the electromagnetic radiation emitted by the Sun is governed by the Stefan-Boltzmann law, which dictates that the total energy radiated per unit surface area of a black body across all wavelengths per unit time is directly proportional to the fourth power of the black body's thermodynamic temperature[5]. The fundamental equation is expressed as:

$$P = \epsilon \sigma A T^4$$

Where σ is the Stefan-Boltzmann constant ($5.670374 \times 10^{-8} \text{ W}/(\text{m}^2 \cdot \text{K}^4)$), ϵ is the emissivity of the object (approximating 1 for the Sun), A is the surface area, and T is the absolute temperature in Kelvin[5]. Based on the Sun's radius ($R_\odot \approx 6.96 \times 10^8 \text{ m}$) and an effective surface temperature of approximately 5778 K, the total luminosity (power output) of the Sun is quantified at 3.846×10^{26} Watts[5].

This radiant energy disperses spherically into a vacuum. At a distance of one astronomical unit ($1 \text{ AU} \approx 1.496 \times 10^{11} \text{ m}$), the mean distance of Earth's orbit, the intensity of this radiation establishes the "solar constant" (S_0). In accordance with the International Astronomical Union (IAU) 2015 Resolution B3, the nominal total solar irradiance is standardized at exactly 1361 W/m^2 [5]. Historical approximations of this value date back to Claude Pouillet's 1838 pyrheliometer measurement of 1228 W/m^2, with modern satellite telemetry confirming the precise value of 1361 W/m^2 during the solar minimum, fluctuating by a maximum of 0.1% during the 11-year solar cycle[9].

Because the Earth's orbit is slightly elliptical, the actual direct solar irradiance measured at the top of the atmosphere (TOA) fluctuates by approximately 6.9% annually, reaching 1412 W/m^2 at perihelion in early January and dropping to 1321 W/m^2 at aphelion in early July[9].

2.2. Geometry of Interception and Global Energy Budget

The total volume of solar energy intercepted by the Earth is a function of the planet's cross-sectional area facing the Sun. Modeling the Earth as a 2D disc with a radius of $R_E = 6.371 \times 10^6 \text{ m}$, the cross-sectional area is:

$$A_{cross} = \pi R_E^2 \approx 1.275 \times 10^{14} \text{ m}^2$$

[cite: 11, 12, 13]

Multiplying this cross-sectional area by the solar constant (S_0) yields the total continuous solar power intercepted by the planet:

$$P_{solar} = \pi R_E^2 \times S_0 = (1.275 \times 10^{14} \text{ m}^2) \times 1361 \text{ W/m}^2 \approx 1.735 \times 10^{17} \text{ W}$$

[cite: 9, 11, 12, 13]

This absolute continuous power input is 1.735×10^{17} Watts, or 173,000 Terawatts (TW)[9].

Due to the Earth's rotation, this energy is distributed across the entire spherical surface area of

the planet ($4\pi R_E^2$). Consequently, the globally and annually averaged solar flux at the top of the atmosphere is exactly one-quarter of the solar constant, equating to roughly $340\ \mathrm{W/m^2}$ [1].

The Earth's energy budget operates on a strict accounting of this incoming $340\ \mathrm{W/m^2}$. Approximately $77\ \mathrm{W/m^2}$ is immediately reflected into space by atmospheric aerosols and clouds, while an additional $23\ \mathrm{W/m^2}$ is reflected by surface albedo[1]. The remaining $240\ \mathrm{W/m^2}$ constitutes the Absorbed Solar Radiation (ASR), which actively drives the planet's climatic, hydrological, and biological systems[1]. Ground-level irradiance on a clear day at the equator during an equinox averages $340\ \mathrm{W/m^2}$ over a 24-hour cycle, dropping to between $550\ \mathrm{W/m^2}$ and $1025\ \mathrm{W/m^2}$ at solar noon depending on cirrus cloud cover[9].

2.3. Quantitative Discrepancy Between Availability and Consumption

The ontological presence of solar energy renders the total sum of all global human energy consumption mathematically insignificant. The physical energy intercepted by the Earth in a single hour is calculated as:

$$E_{1\,\text{hour}} = 1.735 \times 10^{17}\ \text{W} \times 3600\ \text{s} = 6.247 \times 10^{20}\ \text{Joules} = 624.7\ \text{Exajoules (EJ)}$$

[cite: 11, 13]

Extended over a full year, the total solar energy hitting the Earth reaches 5.47×10^{24} Joules, or $5,476,803\ \mathrm{EJ}$, which is equivalent to 173,000,000 Terawatt-hours (TWh)[13].
In stark contrast, the entirety of human primary energy consumption across all sectors—including direct heating, transportation, and industrial processes—averages a continuous draw of 18 TW, equating to approximately 160,000 TWh or 576 EJ per annum[1]. The resulting ratio of annual available solar energy to annual human consumption is precisely 9,508 to 1[13]. Therefore, the Sun delivers more energy to the Earth in exactly 55 minutes than human civilization consumes in an entire year[14].
Furthermore, the combined proven recoverable reserves of all terrestrial fossil fuels and uranium are estimated at approximately 65,000 EJ[14]. The Earth receives this exact amount of solar energy in less than 5 days. Calculations confirm that capturing a mere 1% of the solar energy hitting the planet would yield 17 times the energy required to power current civilization, and covering just 0.3% of the world's land area with photovoltaic infrastructure would fulfill 100% of global electricity demands[14]. The physical reality is an environment characterized by astronomical energy excess.

3. Endogenous Geothermal Dynamics: Planetary Heat

Capacity and Radiogenic Decay

3.1. Quantification of the Internal Heat Budget

The Earth functions as an immense thermal reservoir. The total internal heat content of the planet is quantified at 10^{31} Joules (10^{13} EJ)[4]. When divided by the total annual human energy consumption of 576 EJ, the Earth's internal thermal battery contains enough stored energy to sustain current civilizational energy demands for over 17 billion years, completely independent of solar input[13].

The continuous thermal flux migrating from the Earth's core to its surface operates at a rate of 47 ± 2 TW [1]. Over the course of a standard year, this thermal emission yields between 1,394 EJ and 1,483 EJ of energy, representing more than double the entire primary energy consumption of humanity[4]. The global mean geothermal heat flux is $91.6\ \mathrm{mW/m^2}$, establishing a baseline availability of thermal energy across the entirety of the lithosphere[4].

3.2. Mechanisms of Heat Generation and Convective Transfer

The internal thermal output originates from two primary sources, partitioned in roughly equal proportions. The first is primordial heat—the residual kinetic energy preserved from the planetary accretion process that occurred 4.5 billion years ago, alongside the latent heat of crystallization as the inner core solidifies[4]. The second, and more continuous source, is radiogenic heating, which accounts for approximately 50% of the total internal heat flow[4].

Radiogenic heat is actively generated by the spontaneous radioactive decay of specific isotopes enriched within the mantle and crust. The four dominant heat-producing isotopes are uranium-238 ($^{238}\mathrm{U}$), uranium-235 ($^{235}\mathrm{U}$), thorium-232 ($^{232}\mathrm{Th}$), and potassium-40 ($^{40}\mathrm{K}$)[4]. As these unstable parent isotopes transition to stable daughter configurations, they emit excess internal energy in the form of alpha and beta particles, gamma radiation, and geoneutrinos[4]. The deceleration of these particles within the surrounding crystalline lattice converts kinetic energy directly into sensible heat[20].

This continuous radiogenic and primordial heating elevates the temperature at the core-mantle boundary (2,885 km depth) to over 4000 °C, while the inner core reaches temperatures nearing 6000 °C, analogous to the surface of the Sun[4]. Because the fluidity of the solid mantle is proportional to temperature, this extreme basal heating induces massive mantle convection[4]. Hotter, buoyant mantle material rises toward the lithosphere, while cooler, denser material undergoes subduction[4]. Mantle convection dictates 80% of the Earth's internal heat transport, establishing geothermal gradients that average 25–30 °C per kilometer of depth in standard continental crust[4]. Within the upper 10 kilometers of the contiguous United States alone, the recoverable geothermal energy is estimated at 3.3×10^{25} Joules, a figure that is 6,000 times greater than the energy contained within all known global oil reserves[17].

3.3. Status of Millimeter-Wave Geothermal Drilling (July 2026)

Historically, commercial geothermal energy extraction was spatially confined to low-temperature hydrothermal reservoirs (water and steam) strictly located near tectonic plate boundaries, resulting in a negligible 16 GW of installed global capacity by 2025[16]. However, petrothermal systems (hot dry rock) represent an omnipresent resource accessible globally, provided deep drilling can breach the crystalline basement rock without catastrophic bit wear and prohibitive cost[19].

As of July 2026, the physical barriers to accessing ubiquitous superhot geothermal resources have been definitively nullified by the commercial deployment of millimeter-wave drilling technology. Pioneered by Quaise Energy—a spin-off from the Massachusetts Institute of Technology (MIT)—this drilling architecture replaces mechanical drill bits with a high-power gyrotron[24]. Originally developed for plasma heating in nuclear fusion reactors, the 1-megawatt gyrotron projects focused millimeter-wave electromagnetic energy down a waveguide, directly ablating and vaporizing subterranean basalt and granite[24]. This non-contact technique circumvents the mechanical limits of conventional drilling, requiring no fracking or seismic stimulation[26].

In 2025, Quaise successfully demonstrated the capability to drill over 100 meters through solid granite under full-scale field conditions in Central Texas, and by mid-2026, the company is approaching a continuous depth of 1 kilometer[27]. Having secured $134 million in Series B financing in 2026, the immediate physical trajectory of this technology involves drilling to depths of 20 kilometers within 100-day timeframes, accessing rock temperatures ranging from 300°C to 500°C anywhere on Earth[24].

The primary deployment site, Project Obsidian in Oregon's Deschutes National Forest, represents the world's first commercial superhot geothermal power plant[27]. With construction underway in 2026 and commercial grid operation scheduled for 2030, the facility leverages millimeter-wave technology to generate an initial 50 MW of continuous baseload power, expanding to 250 MW[24]. By utilizing existing fossil-fuel power plant infrastructure and turbines, this technology proves that high-density, 24/7 clean baseload power can be retrofitted globally with a land footprint of less than 1% compared to equivalent solar arrays[24]. The presence of ubiquitous, drillable superhot rock structurally eliminates the concept of geographical energy scarcity.

4. Kinetic and Gravitational Dynamics: Tidal Energy Dissipation

4.1. Orbital Mechanics and Torque Formulation

The third major inexhaustible energy vector is kinetic, derived from the gravitational coupling of the Earth-Moon system. The differential gravitational pull of the Moon (and to a lesser extent, the Sun) across the diameter of the Earth generates tidal forces, distorting the oceans and the

solid Earth into prolate ellipsoids with tidal bulges[31]. Because the Earth's sidereal rotation period (approximately 24 hours) is significantly faster than the Moon's orbital period (approximately 27.3 days), the Earth's surface rotates "through" these tidal bulges[15].
However, the response of the oceans and the solid Earth is not instantaneous. Friction and the dissipation of energy within the water column cause a lunitidal interval, meaning the tidal bulge is carried forward by the Earth's rotation, remaining slightly ahead of the direct Earth-Moon axis[32]. This misalignment induces a continuous gravitational torque between the offset bulge and the Moon[32]. According to the laws of conservation of angular momentum and energy, this torque transfers angular momentum from the Earth's rotation to the Moon's orbit[32]. Consequently, the Moon recedes from the Earth at a measured rate of $38.30 \pm 0.08 \text{ mm/year}$, while the Earth's rotation decelerates, increasing the length of the day by approximately 1.5 milliseconds per century[15].

4.2. Quantification of Tidal Energy Dissipation

The continuous reduction in the Earth's rotational velocity acts as an immense planetary flywheel discharging kinetic energy into the oceans. The rotational kinetic energy (E) of the Earth is defined as:

$$E = \frac{1}{2} I \omega^2$$

Where I is the Earth's moment of inertia ($8.04 \times 10^{37} \text{ kg} \cdot \text{m}^2$) and ω is the angular velocity ($7.292 \times 10^{-5} \text{ rad/s}$)[15]. The derivative of this energy with respect to time gives the exact power dissipation:

$$\frac{dE}{dt} = I \omega \frac{d\omega}{dt}$$

Given the observed lengthening of the day, the angular deceleration ($d\omega/dt$) is calculated at $-4.0 \times 10^{-22} \text{ rad/s}^2$[15]. The total power stripped from the Earth's rotation is therefore:

$$P = (8.04 \times 10^{37}) \times (7.292 \times 10^{-5}) \times (-4.0 \times 10^{-22}) \approx -2.34 \times 10^{12} \text{ W} = 2.34 \text{ TW}$$

[cite: 15, 33]
When accounting for lunar internal dissipation, core-mantle boundary friction, and solar tides, the total accepted astronomical rate of tidal energy dissipation in the Earth-Moon system is precisely 3.7 Terawatts (TW)[32].
Approximately 2.7 TW to 3.1 TW of this energy is dissipated as frictional heat in the turbulent bottom boundary layers of shallow marginal seas, such as the Patagonian Shelf[31]. High-level

satellite altimetry (TOPEX/Poseidon) confirms that the remaining 1 TW of dissipation occurs in the deep ocean via topographic scattering of tidal energy into internal waves, a process critical for maintaining the large-scale "conveyor-belt" abyssal circulation[31]. This 3.7 TW represents a permanent, predictable mechanical energy flux that is fundamentally immune to atmospheric weather patterns or climatological shifts, available for localized capture via tidal barrage systems (e.g., the 254 MW Sihwa Lake facility)[15].

5. Lithospheric Material Composition and the Thermodynamics of Infinite Recyclability

5.1. Crustal Elemental Abundances and Absolute Availability

While energy behaves as an open flow, planetary matter is conserved. The concept of material scarcity is frequently asserted in economic models; however, geological sampling and spectrographic data confirm that the Earth's crust is overwhelmingly abundant in the chemical elements required for an advanced industrial civilization.

The crustal abundance of elements refers to the average mass fraction of elements within the planet's outermost solid layer. More than 99% of the mass of the Earth's crust is composed of just a few dozen minerals, and over 74% of the crust is comprised of just two elements: oxygen and silicon (forming various silicate minerals like quartz and feldspar)[36]. The precise compositional breakdown of the most abundant elements by weight percent and parts per million (ppm) is as follows:

Atomic Number	Element	Symbol	Classification	Mass Fraction (ppm)	Weight Percent (%)
8	Oxygen	O	Lithophile	461,000	46.10%
14	Silicon	Si	Lithophile	282,000	28.20%
13	Aluminum	Al	Lithophile	82,300	8.23%
26	Iron	Fe	Siderophile	56,300	5.63%
20	Calcium	Ca	Lithophile	41,500	4.15%
11	Sodium	Na	Lithophile	23,600	2.36%
12	Magnesium	Mg	Lithophile	23,300	2.33%

19	Potassium	K	Lithophile	20,900	2.09%
22	Titanium	Ti	Lithophile	5,650	0.565%
1	Hydrogen	H	Atmophile	1,400	0.140%
15	Phosphorus	P	Lithophile	1,050	0.105%
25	Manganese	Mn	Lithophile	950	0.095%
6	Carbon	C	Atmophile	200	0.020%

[36]

This data demonstrates that industrial cornerstones such as aluminum (8.23%) and iron (5.63%) are the third and fourth most abundant elements in the crust, available in quantities numbering in the billions of metric tons[38]. Variations exist between the upper continental crust (andesitic/tonalitic, with higher SiO_2 content around 66.6%) and the oceanic crust (mafic, higher iron and magnesium content), yet the sheer elemental mass precludes any physical possibility of absolute exhaustion[36].

5.2. The Law of Conservation of Mass and Thermodynamic Limits

The concept of "running out of materials" violates the fundamental physical laws governing the universe. The First Law of Thermodynamics (the principle of mass-energy conservation) strictly dictates that matter can neither be created nor destroyed[2]. When industrial processes manufacture a good, or when a product is discarded into a landfill, the constituent atoms of copper, steel, or aluminum do not vanish; they are merely redistributed into different chemical bonds or dispersed spatially[3].

The physical barrier to total material circularity is defined solely by the Second Law of Thermodynamics, which dictates that the entropy of a closed system always increases[44]. In economic-ecological terms (as theorized by Nicholas Georgescu-Roegen and Robert Ayres), industrial processes take low-entropy, highly concentrated mineral deposits and transform them into high-entropy, highly dispersed waste[44]. Separating these dispersed elements (e.g., removing tramp elements like zinc or magnesium from recycled automotive aluminum alloys, or recovering rare earth metals from molecularly complex printed circuit boards) requires the input of work to reverse the entropy of mixing[43].

Therefore, the thermodynamic limit to recycling is not the availability of the material itself, but the availability of the *energy* required to re-concentrate the material[2].

Because the Earth is an open thermodynamic system with respect to energy—receiving a

continuous 1.73×10^{17} W from the Sun and 47 TW from geothermal sources—the thermodynamic penalty for reversing entropy is entirely negated[4]. Metals possess the innate physical property of "infinite recyclability"[51]. When sufficient energy is applied to metallurgical recycling (via electric arc furnaces, reverse Knudsen pyrometallurgy, or solid-state electrolysis), the recovered metallic elements are chemically and physically indistinguishable from virgin mined ores[43]. Furthermore, recycling generally demands an order of magnitude less energy than primary extraction; processing secondary copper requires 42% to 95% less energy than virgin ore (dropping from 58 GJ/tonne to 2.9-33.8 GJ/tonne), and secondary aluminum requires merely 5% of the energy needed for primary smelting[43]. Consequently, in the presence of limitless solar and geothermal power, material scarcity is a physical impossibility.

6. The Architecture of Artificial Scarcity and "Pecuniary Sabotage"

If physics confirms that energy is inexhaustible and matter is infinitely recyclable, the prevailing condition of economic limitation must be understood as an engineered structural phenomenon. This is classified as "artificial scarcity"—the deliberate, human-imposed restriction on the production, distribution, replication, or availability of goods, services, or information that could otherwise be provided in abundance at near-zero marginal cost[54].

6.1. Veblenian Mechanics of the Business Enterprise

The structural foundation of artificial scarcity was rigorously formulated by institutional economist Thorstein Veblen in his 1904 treatise, *The Theory of Business Enterprise*[56]. Veblen differentiated between the "machine process" of industry (driven by engineers and technicians seeking maximum technical efficiency, continuity, and output) and the "pecuniary discipline" of business (driven by absentee owners and financiers seeking maximum monetary profit)[56].
Veblen identified that in a highly productive industrial system, allowing the unhindered output of goods would result in overproduction and a subsequent collapse of prices, destroying profitability[56]. Therefore, the modern business enterprise is structurally incentivized to prioritize "vendibility" over "serviceability"[56]. To maintain prices, businesses must continuously engage in what Veblen termed "pecuniary sabotage" or the "conscientious withdrawal of efficiency"[56]. Sabotage, in this economic context, is the deliberate, legal practice of restricting, delaying, or obstructing production to secure a special advantage[58].
Rather than a breakdown of the market system, artificial scarcity is the intentional regulatory mechanism of it[58]. It converts non-rivalrous abundance into excludable assets, effectively decoupling economic value from the actual physical cost of production, which approaches zero in automated and digital systems[55].

6.2. Modalities of Imposed Scarcity

The implementation of pecuniary sabotage and artificial scarcity manifests through several highly documented empirical practices:

1. **Intellectual Property and Digital Rights Management (DRM):** In the digital and informational realms, the marginal cost of reproducing data, software, and biological sequences is practically zero[54]. Patents and copyrights act as statutory monopolies, granting exclusive rights to withhold knowledge[54]. DRM algorithms criminally prohibit the otherwise costless replication of software, artificially enforcing rivalry onto non-rivalrous digital goods to maintain subscription rents[54].
2. **Planned Obsolescence:** Engineering products with deliberately shortened life spans, non-modular architectures, or firmware blocks (e.g., software updates that degrade battery performance on older devices) to force continuous cyclical consumption and artificial material throughput[62].
3. **Destruction of Surplus:** The literal destruction of manufactured inventory to prevent market saturation. Luxury fashion brands, electronic retailers, and agricultural cartels frequently incinerate, slash, or dispose of millions of dollars of perfectly functional excess stock to artificially maintain price floors and brand exclusivity[54].
4. **Spatial and Real Estate Financialization:** The enclosure of common land and the financialization of housing, where real estate is treated as a speculative asset rather than a utility[66]. The hoarding of vacant properties and deliberate restriction of supply drives up land rent and housing costs, manufacturing a shortage in cities with vast, unused architectural footprints[65].
5. **Agricultural Demand Management:** To combat the "treadmill" of massive agricultural overproduction caused by technological yields, policies mandate the conversion of edible crops into biofuels, or pay farmers to leave arable land fallow, actively suppressing food availability to stabilize commodity prices[54].

6.3. The Cognitive "Hegemony of Scarcity"

The persistent enforcement of these mechanisms yields profound socio-psychological consequences. Extensive research by behavioral economists Eldar Shafir and Sendhil Mullainathan demonstrates that operating within an environment of manufactured resource restriction induces a "scarcity mindset"[67]. This phenomenon causes cognitive tunneling—the brain forcefully concentrates on immediate deficits (lack of time, money, or food) and expends immense "mental bandwidth" simply managing daily survival[67].

This depletion of cognitive capacity impairs executive functioning, lowers long-term strategic planning abilities, and heightens stress, effectively trapping populations in a self-reinforcing cycle of reactionary, short-term decision-making (firefighting)[67]. Scarcity is therefore not merely an economic condition, but a socially constructed epistemological framework (the "social construction of scarcity"), where the academic discourse of economics actively normalizes deprivation by defining human existence solely through the lens of limited resources, thereby obscuring the true social origins of poverty[71].

7. Conclusion: The Ontological Definition of

Subservience Before Artificial Scarcity

The empirical synthesis of planetary physics and institutional economics demands a strict re-evaluation of the human condition as of July 2026. The Earth is demonstrably bombarded with 1.73×10^{17} Watts of solar energy, continuously boils with 10^{31} Joules of geothermal heat available globally via millimeter-wave ablation, generates 3.7 Terawatts of tidal momentum, and contains a lithosphere saturated with infinitely recyclable metals governed by the absolute conservation of mass.

In the presence of this irrefutable physical abundance, the ongoing human experience of limitation requires definitive categorization.

Based strictly on the data, the **subservience (присмыкание) of humanity before self-created artificial scarcity and lack of resources is defined as the voluntary cognitive, institutional, and socio-economic subordination of the species to a mathematically and physically fictitious paradigm of limitation.**

It is the systemic capitulation to "pecuniary sabotage" and excludability—a condition wherein the species willfully suppresses its own technological output, actively destroys its own surplus, and enforces cognitive tunneling upon its own population solely to sustain the mechanisms of monopoly rent, price maintenance, and absentee ownership. This subservience represents the ultimate ontological paradox of the era: a civilization choosing to simulate poverty, starvation, and strict resource rationing within the physical boundaries of a thermodynamic engine capable of generating absolute, inexhaustible abundance.

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